

Another note to take is that it is not necessary to have such a highly symmetric cluster Hamiltonian (2). The mappings to pseudo-spin-1/2 should work as long as the ground states of the cluster Hamiltonian are the two-fold degenerate singlets. One generalization, which conforms the symmetry of the lattice in FIG. 2, is to have

$$H_{\text{cluster}} = (J_{\text{cluster}}/2)(r \cdot \mathbf{S}_1 + \mathbf{S}_2 + \mathbf{S}_3 + \mathbf{S}_4)^2 \quad (11)$$

with $J_{\text{cluster}} > 0$ and $0 < r < 3$. However this is not convenient for later discussions and will not be used.

We briefly describe some of the properties of (8). Its low energy states are entirely in the space that each of the clusters is a physical spin singlet (called cluster singlet subspace hereafter). Therefore physical spin correlations are strictly confined within each cluster. The excitations carrying physical spin are gapped, and their dynamics are ‘trivial’ in the sense that they do not move from one cluster to another. But there are non-trivial low energy physical spin singlet excitations, described by the pseudo-spins defined above. The correlations of the pseudo-spins can be mapped to correlations of their corresponding physical spin observables (the inverse mappings are not unique, c.f. TABLE I). For example $\tau^{x,y}$ correlations become certain dimer-dimer correlations, τ^z correlation becomes chirality-chirality correlation, or four-dimer correlation. It will be interesting to see the corresponding picture of the exotic excitations in the Kitaev model, e.g. the Majorana fermion and the Ising vortex. However this will be deferred to future studies.

It is tempting to call this as an exactly solved spin liquid with spin gap ($\sim J_{\text{cluster}}$), an extremely short-range resonating valence bond(RVB) state, from a model with spin rotation and time reversal symmetry. However it should be noted that the unit cell of this model contains an even number of spin-1/2 moments (so does the original Kitaev model) which does not satisfy the stringent definition of spin liquid requiring odd number of electrons per unit cell. Several parent Hamiltonians of spin liquids have already been constructed. See for example, Ref.^{24–27}.

IV. GENERATE THE HIGH ORDER PHYSICAL SPIN INTERACTIONS BY PERTURBATIVE EXPANSION.

One major drawback of the present construction is that it involves high order interactions of physical spins[see (8) and (9)], thus is ‘unnatural’. In this Section we will make compromises between exact solvability and naturalness. We consider two clusters j and k and try to generate the $J_{x,y,z}$ interactions in (7) from perturbation series expansion of more natural(lower order) physical spin interactions. Two different approaches for this purpose will be laid out in the following two Subsections. In Subsection IV A we will consider the two clusters as two tetrahedra, and couple the spin system to certain optical phonons, further coupling between the phonon modes

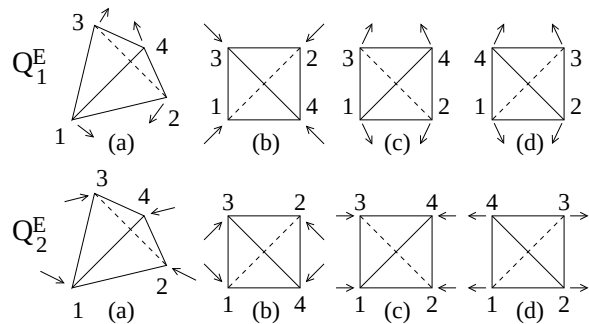


FIG. 3: Illustration of the tetragonal to orthorhombic Q_1^E (top) and Q_2^E (bottom) distortion modes. (a) Perspective view of the tetrahedron. 1, ..., 4 label the spins. Arrows indicate the motion of each spin under the distortion mode. (b) Top view of (a). (c)(d) Side view of (a).

of the two clusters can generate at lowest order the desired high order spin interactions. In Subsection IV B we will introduce certain magnetic, e.g. Heisenberg-type, interactions between physical spins of different clusters, at lowest order(second order) of perturbation theory the desired high order spin interactions can be achieved. These approaches involve truncation errors in the perturbation series, thus the mapping to low energy effect Hamiltonian will no longer be exact. However the error introduced may be controlled by small expansion parameters. In this Section we denote the physical spins on cluster $j(k)$ as $j1, \dots, j4$ ($k1, \dots, k4$), and denote pseudo-spins on cluster $j(k)$ as $\tilde{\tau}_j(\tilde{\tau}_k)$.

A. Generate the High Order Terms by Coupling to Optical Phonon.

In this Subsection we regard each four-spin cluster as a tetrahedron, and consider possible optical phonon modes(distortions) and their couplings to the spin system. The basic idea is that the intra-cluster Heisenberg coupling J_{cluster} can linearly depend on the distance between physical spins. Therefore certain distortions of the tetrahedron couple to certain linear combinations of $\mathbf{S}_\ell \cdot \mathbf{S}_m$. Integrating out phonon modes will then generate high order spin interactions. This idea has been extensively studied and applied to several magnetic materials^{28–34}. More details can be found in a recent review by Tchernyshyov and Chern³⁵. And we will frequently use their notations. In this Subsection we will use the representation (5) for τ^z .

Consider first a single tetrahedron with four spins 1, ..., 4. The general distortions of this tetrahedron can be classified by their symmetry (see for example Ref.³⁵). Only two tetragonal to orthorhombic distortion modes, Q_1^E and Q_2^E (illustrated in FIG. 3), couple to the pseudo-spins defined in Section II. A complete analysis of all modes is given in Appendix A. The coupling is of the